M Saim Ahmad(23i-0781)
M Ismail(23i-2524)
CS-B(OS)

Rubrics for AirControlX Project Evaluation

Total Marks: 350

1. Module 1: System Rules & Restrictions (100 Marks)

Criteria	Excellent (90–100%)	Good (70–89%)	Fair (50–69%)	Poor (<50%)
Aircraft & Airline Restrictions (20 marks)	Correctly implements all 6 airlines, aircraft types, and flight counts. Handles cargo/emergency restrictions flawlessly.	Minor errors in airline/flight counts. Cargo/emergency logic mostly correct.	Significant errors in airline/flight counts. Partial cargo/emergency logic.	Fails to implement airlines or aircraft types.
Runway Allocation (20 marks)	Runways assigned per direction and type (A/B/C). Strict synchronization (mutex/semaphores) for single-use enforcement.	Minor issues in direction-based allocation or synchronization.	Runway assignment logic inconsistent. Weak synchronization.	No synchronization or incorrect runway assignments.
Speed Monitoring & AVN (30 marks)	Full compliance with speed rules for all phases (arrival/departure). AVN triggers correctly for violations.	Minor speed rule deviations (e.g., 1–2 phases incorrect). AVN activation mostly works.	Multiple speed rule violations. AVN logic partially missing.	No speed monitoring or AVN implementation.
Simulation Time & Fault Handling (15 marks)	5-minute simulation runs smoothly. Ground faults handled (e.g., towing).	Simulation runs but timing issues. Basic fault handling.	Simulation incomplete. Fault handling missing.	No simulation or fault handling.
Code Structure (15 marks)	Clean, modular code with classes. Proper use of C++ and OS concepts.	Readable code but minor structural issues.	Poorly structured code. Minimal OOP.	Unreadable or non-functional code.

2. Module 2: ATCS Core Functionalities (130 Marks)

Criteria	Excellent (90–100%)	Good (70–89%)	Fair (50–69%)	Poor (<50%)
Flight Scheduling & Conflict Resolution (25 marks)	Flights added dynamically. Conflict resolution (priority/FCFS) works perfectly.	Minor scheduling errors. Basic conflict handling.	Incomplete scheduling. Weak conflict logic.	No scheduling or conflict resolution.
Runway Allocation & Sync (25 marks)	Runway allocation aligns with direction/priority. Thread synchronization robust.	Minor allocation/sync issues (e.g., rare race conditions).	Frequent sync errors. Allocation logic flawed.	No synchronization or allocation logic.
Phase Management (30 marks)	Seamless transitions (holding → landing, gate → departure). Realistic timing.	Phase transitions mostly correct. Minor timing issues.	Incomplete transitions (e.g., missing phases).	No phase management.
Priority Handling (25 marks)	Emergency flights pre-empt queues. RWY-C used for overflow.	Minor priority errors (e.g., delays in emergency handling).	Partial priority logic. Overflow handling missing.	No priority handling.
Queue & Status Tracking (25 marks)	Real-time queue updates. Accurate status tracking (e.g., taxiing, cruising).	Minor delays in updates. Status tracking mostly correct.	Inconsistent updates. Partial tracking.	No queue/status tracking.

3. Module 3: Subsystems & Airline Integration (120 Marks)

Criteria	Excellent (90–100%)	Good (70–89%)	Fair (50–69%)	Poor (<50%)
ATCS Controller & Violation Detection (30 marks)	Real-time monitoring of speed/altitude. AVN activation accurate.	Minor detection delays. AVN activation mostly correct.	Detection logic flawed. Partial AVN handling.	No violation detection.
AVN Generator & Payment (30 marks)	AVN details (fine calculation, service fee) fully implemented. Payment confirmation works.	Minor errors in fine calculation. Payment flow mostly works.	AVN details incomplete. Payment logic missing.	No AVN/payment system.
Airline Portal & StripePay (25 marks)	Portal displays AVNs and payment status. StripePay integration functional.	Minor UI issues. Payment flow partially works.	Portal incomplete. StripePay non-functional.	No portal/payment integration.
Simulation & Visualization (35 marks)	Smooth animations (SFML/OpenGL). Color-coded runways/aircraft. Real-time status updates.	Basic graphics with minor lag. Partial visual elements.	Static/no animations. Minimal visualization.	No graphical output.

4. General Evaluation (Weighted Across Modules)

Criteria	Marks	Description
Code Quality	20	Clean, efficient, and well-documented code. Proper use of OS concepts (threading, IPC).
Project Report & Self-Assessment	15	Detailed report explaining design choices. Honest self-assessment form.
Timeliness	Penalties	-20% per module for late submission (applied to module marks).
Plagiarism	Zero	Immediate disqualification if detected.
Viva Performance	30	Clear understanding of OS concepts. Ability to justify design decisions and debug issues.

Penalties

- **Late Submission**: -20% per module.
- **Plagiarism**: Zero marks for all group members.
- **Incomplete Submission**: Partial marks based on missing components.